

Qatar-1b

R-1587

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_211031 · created 2026-08-02T00:35:24+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459518.6159 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.031
Transit depth (Rp/Rs)^2	2.19 +/- 0.93 [%]
Orbital Inclination (inc)	89.06 +/- 2.1
Airmass coefficient 1 (a1)	1.253 +/- 0.019
Airmass coefficient 2 (a2)	-0.165 +/- 0.048
Scatter in the residuals of the lightcurve fit is	1.54 %
Best Comparison Star	None
Optimal Aperture	4.16
Optimal Annulus	6.58
Transit Duration (day)	0.0788 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.031 vs 0.1463 (deviation 0.04 σ)
Duration fit vs published	0.0788 d vs 0.0692 d (deviation 2.32 σ)
Mid-time O-C	-5.4 min
Depth SNR	3.8
Residual scatter	1.54 %

Light curve

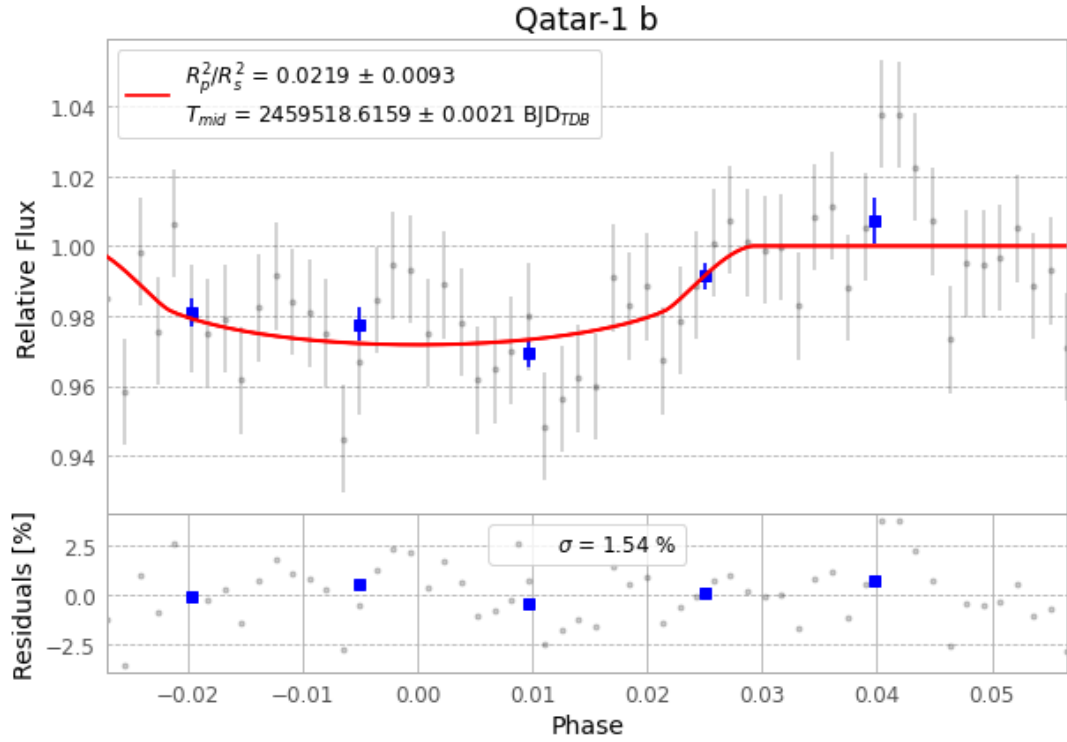


Figure 1 — FinalLightCurve_Qatar-1 b_2021-10-30.png (SHA-256 dd56145cd1480cda...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 243], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 547186b11440d18d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

58 FITS frames (38 MB), Qatar-1211031014812.FITS → Qatar-1211031043912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-211031.log (ba419673ca85...), FinalLightCurve_Qatar-1 b_2021-10-30.png (dd56145cd148...), FinalLightCurve_Qatar-1 b_2021-10-30.csv (169faf81bc2a...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 00:35Z.